Last time, we studied the simple linear regression model

$$y = \beta_0 + \beta_1 x + \varepsilon$$

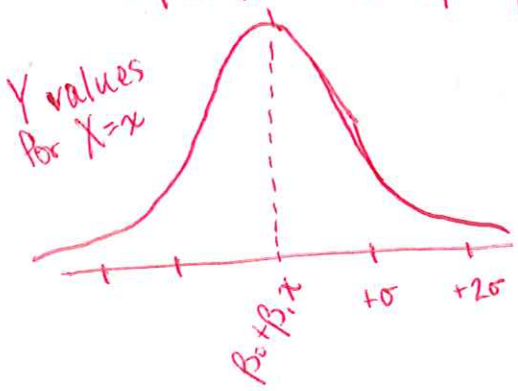
under the additional assumption that the error terms (the ε's) are independent, and normally distributed with variance σ^2 (and still expected value of 0, like before).

new assumption: $\varepsilon_1, \varepsilon_2, \varepsilon_3, \ldots \overset{iid}{\sim} N(0, \sigma^2)$

Assuming the error terms are normally distributed had several useful consequences.

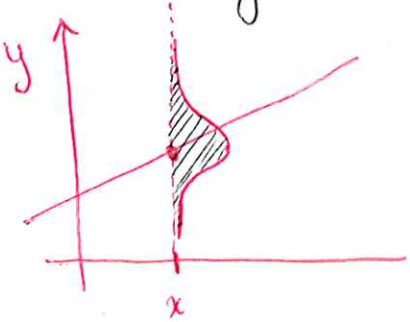
Consequences

① $Y|X=x \sim N(\beta_0+\beta_1 x, \sigma^2)$



First, for any particular value of the predictor variable X, the response variable Y is a random variable that also follows a normal distribution.

If we collected data we would see a variety of different y values for observations with that particular x value. But overall, those y values would be centered at the point on the line $(x, \beta_0+\beta_1 x)$, and would be spread out above and below the regression line according to a normal distribution.



Another consequence of the error terms following a normal dist. is that the least squares estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are also normally distributed.

② $\hat{\beta}_0, \hat{\beta}_1$ also normal

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{(n-1)S_x^2}\right)$$

sample variance $S_x^2 = \frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{n-1}$

Because the least squares estimator of the slope, $\hat{\beta}_1$, is normally distributed, we can determine the probability of getting an estimated slope within a certain number of standard deviations of the true value.

$$P(\hat{\beta}_1 \text{ within } \pm 2 \text{ sd of } \beta_1) \approx 95\%$$

(note: the z score, $\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{\sigma^2}{(n-1)S_x^2}}}$ follows a $N(0,1)$ distribution)

However, the "standard deviation" in that statement refers to the standard deviation of $\hat{\beta}_1$, which depends on σ^2:

$$sd = \sqrt{\frac{\sigma^2}{(n-1)S_x^2}} \rightarrow \text{unknown}$$

σ^2 is the variance of the error terms. It describes how vertically spread out the y values are for any particular x value, in the true model for the population. We don't know σ^2, but we can estimate it using the residuals from the fitted model.

$$\varepsilon_i = y_i - (\beta_0 + \beta_1 x_i) \quad \text{(error term)}$$

$$\hat{e}_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i) \quad \text{(residual)}$$

estimate $\hat{\sigma}^2$ = sample var of residuals

Specifically, we can use the sample variance of the residuals as an estimate of σ^2.

Recall that when we constructed confidence intervals and performed hypothesis tests about a population mean μ, if we estimated σ^2 from the data, we based our probability calculations on the t distribution instead of the normal distribution.

recall: $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$

inference for μ used t dist (not normal) when σ unknown

Because we are estimating the variance of the error terms, σ^2, from the data in the linear regression setting, we will once again base our probability calculations on a t dist when performing inference about the slope β_1.

inference about β_1 also uses t dist (σ^2 unknown)

In the case of a linear regression model with one predictor variable x, the "degrees of freedom" for the t distribution is $n-2$.

df = $n-2$ now

This will be our basis for constructing confidence intervals and performing hypothesis tests about the true slope β_1 in the population model.

Confidence intervals

Suppose we are interested in the language acquisition study from previous lectures.

ex: Y = # words spoken to infant daily (in 1000's)

Y = vocab size (# words) @ age 24mo

observe $(x_1, y_1), \ldots, (x_{30}, y_{30})$

we observe 30 children, and fit a least squares regression

line to their data.

estimate $\hat{y} = 158 + 16.9x$ $\quad \hat{\beta}_1 = 16.9$

We estimate that for children whose infant-directed speech differs by 1000 words, the infant hearing more words has a vocabulary size 16.9 words larger, on average. But we don't think the average difference is EXACTLY 16.9 words - that's just the estimate we obtained from this particular sample of 30 children.

what is plausible range of values for β_1?

If we observed a different 30 babies, we'd get a slightly different slope. The true value for the entire population could be higher or lower than 16.9. Based on our observed data, what would be a plausible range for β_1?

To construct a confidence interval for β_1, we need to calculate $\hat{\sigma}^2$ and S_x^2 from the data. (In practice, this isn't something we will calculate manually.)

suppose we calculate $\hat{\sigma}^2 = 2004$ sample var of residuals $\quad S_x^2 = 8.84$ sample var of x $\quad$ from the data

Across different random sample data sets, the estimated slope is within ± 2 standard deviations of the true value (β_1) 95% of the time. IF we knew σ^2 (and, thus, the standard deviation of $\hat{\beta}_1$), we could construct a 95% CI for β_1 by going 2sd from $\hat{\beta}_1$ in either direction.

normal dist: $\hat{\beta}_1, \beta_1$ within ± 2 sd of each other 95% of the time

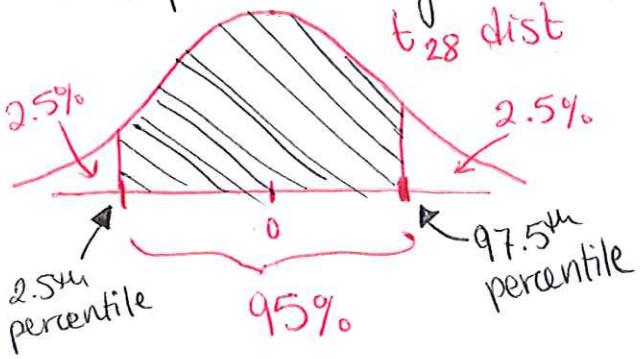
95% CI for β_1: $\hat{\beta}_1 \pm 2$ sd (if sd KNOWN)

Under a normal distribution, ± 2 standard deviations from the mean captures 95% of the total probability.

In our case, because the standard deviation is estimated from the data, we use the t distribution instead.

sd estimated from data → use t dist.

For a 95% confidence interval, the relevant question is: how far from the mean must we go to capture 95% of the probability, under a t distribution w/ n-2 degrees of freedom?



Looking up the 97.5th percentile of a t dist with df = 28, (n=30 in our example) we get:

$qt(0.975, df=28) \approx 2.05$

(The t distribution is symmetric, so the 2.5th percentile is -2.05.)

So, to construct our 95% CI for β_1, we go 2.05 standard deviations in either direction from the estimated slope $\hat{\beta}_1$,

$$95\%\ CI \text{ for } \beta_1 : \left[\hat{\beta}_1 - 2.05\ se(\hat{\beta}_1),\ \hat{\beta}_1 + 2.05\ se(\hat{\beta}_1)\right]$$

$$se(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{(n-1)S_x^2}}$$

where the "standard deviation" uses the $\hat{\sigma}^2$ value estimated from the data. (We sometimes refer to this as the standard error of the estimator $\hat{\beta}_1$, aka $se(\hat{\beta}_1)$.)

If we want a different level confidence interval instead of 95%, it follows the same form - we just check how many standard deviations we need to go in each direction to get to the desired probability under a t distribution with $n-2$ degrees of freedom (by looking up the appropriate percentile of that t distribution).

general for any level CI:

$$\text{Estimate} \pm [\text{percentile of t dist}] * [\text{se(estimator)}]$$

Let's plug in the values to find the 95% CI for our current example:

$$16.9 \pm 2.05 \sqrt{\frac{2004}{(30-1)(8.84)}}$$

$\hat{\beta}_1$; t_{n-2} dist 97.5th %tile ; $\hat{\sigma}^2$; $n-1$; s_x^2

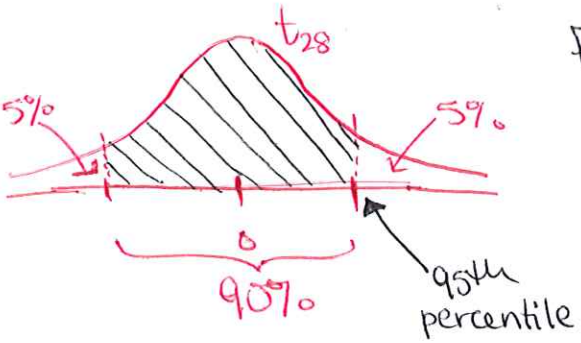
$16.9 \pm 5.7 \rightarrow 11.2$ to 22.6

95% confident true slope btwn 11.2 and 22.6

To summarize this result, we would say that we are "95% confident" the true avg. diff in 24mo vocabulary between babies hearing a difference of 1,000 infant-directed words daily is between 11.2 and 22.6 words.

Next, let's find a 90% confidence interval for β_1 instead.

90% CI for β_1



For a 90% interval, we need to find the 95th percentile.

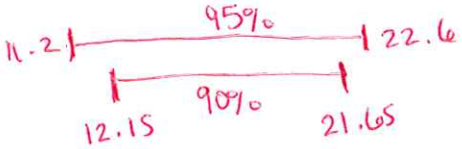
$qt(0.95, df = 28) \approx 1.7$

Plugging in 1.7 as the percentile (in other words, the # of standard deviations to go in either direction to construct our confidence interval), we get:

$$\hat{\beta}_1 \pm 1.7 \sqrt{\frac{2004}{(30-1)8.84}}$$

$$16.9 \pm 4.75$$

$$\approx 12.15 \text{ to } 21.65$$



Just like we saw with confidence intervals for proportions and means, a higher confidence "level" corresponds to a wider interval.

higher conf level → wider interval

In order to be "95% confident", we need to include a larger range of possible values, compared to only being "90% confident".

When we assume the error terms (the ε's) are iid normal, the least squares estimator of the intercept is also normally distributed.

intercept $\quad \varepsilon_1, \varepsilon_2, \varepsilon_3, \ldots \overset{iid}{\sim} N(0, \sigma^2) \Rightarrow \hat{\beta}_0$ normal

$$\hat{\beta}_0 \sim N\left(\beta_0, \underbrace{\sigma^2\left(\frac{1}{n} + \frac{\bar{X}^2}{(n-1)S_x^2}\right)}_{\text{var}(\hat{\beta}_0)}\right)$$

you are not required to know this variance expression

Just like we saw for the slope, the variance of the intercept estimator also depends on σ^2 (the variance of the error terms).

$\text{Var}(\hat{\beta}_0)$ depends on σ^2

↳ σ^2 unknown, use $\hat{\sigma}^2$ from residuals

And once again, we don't know σ^2, so we use an estimate calculated from the data (sample variance of the residuals). Using an estimate for σ^2 means we also use the t distribution for inference about the true intercept β_0, and once again the degrees of freedom value is $n-2$.

use t_{n-2} dist for inference

To construct confidence intervals for the intercept, we follow the same format as for the slope:

CI for true β_0: $\hat{\beta}_0 \pm \left[\begin{array}{c}\text{percentile} \\ \text{of } t_{n-2} \\ \text{dist.}\end{array}\right] * SE(\hat{\beta}_0)$

↖ ie $\sqrt{\hat{\sigma}^2_\varepsilon\left(\frac{1}{n} + \frac{\bar{x}^2}{(n-1)s_x^2}\right)}$

Rather than constructing confidence intervals manually, we can also obtain them in R using the confint() function.

~ examples in notebook ~

Under the assumption that the random error terms are iid normal, we can also perform hypothesis tests about the true value of the slope or intercept.

$\varepsilon_1, \varepsilon_2, \varepsilon_3, \ldots \overset{iid}{\sim} N(0, \sigma^2)$ → $\hat{\beta}_1 \sim N(\beta_1, \text{var}(\hat{\beta}_1))$

→ allows hypothesis tests

For a particular hypothesized value of β_1, we can check whether the estimated $\hat{\beta}_1$ is too far away (too many standard deviations) from the hypothesized value for the hypothesized value to actually be plausible. In particular, we often want to test the null hypothesis that the slope equals zero.

often want to test $H_0: \beta_1 = 0$ vs. $H_A: \beta_1 \neq 0$

If the slope equals zero, that means there is no linear relationship between X and Y.

$\beta_1 = 0$ → no linear assoc. btwn X and Y

$Y = \beta_0 + \varepsilon$

If the slope is zero, then Y values follow the same distribution ($N(\beta_0, \sigma^2)$) regardless of the value of X. In other words, knowing the value of X would not help us predict the value of Y (using a linear model).

* linear model: knowing x_i doesn't help predict y_i

$Y \sim N(\beta_0, \sigma^2)$ for ANY X

If the true slope is zero, and we estimate the slope from a random sample of data, our $\hat{\beta}_1$ estimate won't be exactly zero — a line with a nonzero slope might provide a slightly better fit to the set of sample data points we happened to get, just due to random chance.

if $\beta_1 = 0$, estimated $\hat{\beta}_1$ won't exactly $= 0$
due to randomness
↳ how FAR from 0 is it likely to be?

But how far away from zero is it likely to be? We can calculate this for the language acquisition data set:

language data: $\hat{\beta}_1 = 16.9$

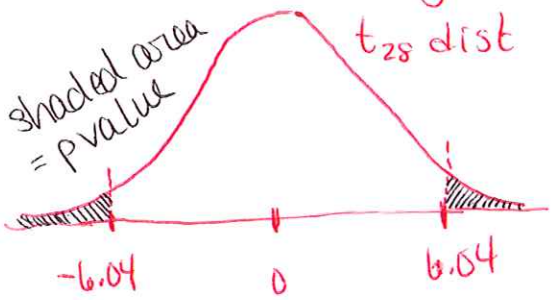
$$\frac{16.9 - 0}{\sqrt{\frac{\hat{\sigma}^2}{(n-1)S_x^2}}} \approx \frac{16.9}{\sqrt{\frac{2004}{(30-1)(8.84)}}} \approx 6.04$$

The estimated slope of $\hat{\beta}_1 = 16.9$ is roughly 6.04 standard deviations away from zero (where the size of a standard deviation was estimated using the observed data).

If the null hypothesis is true, i.e. $\beta_1 = 0$, what is the probability of getting a value at least this far away from zero, under a t distribution with $n-2 = 28$ degrees of freedom?

$H_0: \beta_1 = 0$, $H_A: \beta_1 \neq 0$, $\alpha = 0.05$

$\mathbb{P}(\geq 6.04$ sd away from $0) = ?$



lower tail probability (the two tails are symmetric)

$2 * pt(-6.04, df = 28)$

$\approx 1.64 \times 10^{-6}$

p value

If the slope truly were zero, the probability of getting an estimated slope at least this far away from zero is approximately 1.64×10^{-6}.

i.e. prob. of observing data w/ $\hat{\beta}_1$ at least this far from 0 (the expected value under H_0)

That probability is quite small. With a rejection threshold of $\alpha = 0.05$, we would reject H_0, and conclude that there is evidence of a nonzero linear relationship between X and Y.

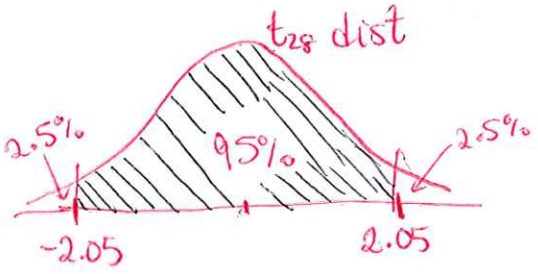
p value $< \alpha = 0.05$ → reject H_0, conclude there is linear assoc btwn X and Y

We can also reach this conclusion without explicitly finding a p value. When we constructed a confidence interval earlier, we looked up the 97.5th percentile and found that the estimate falls within 2.05 standard deviations 95% of the time, under a t distribution with 28 degrees of freedom. So if our estimated slope of 16.9 is 6.04 standard deviations away, the probability of a result like this must be less than 5%.

alternative to p value: critical value

We can determine how many sd away from the expected value (ie $\beta_1 = 0$) our estimated $\hat{\beta}_1$ would have to be in order for us to reject the null hypothesis. That cutoff is called a "critical value".

for $\alpha = 0.05$, how many sd away causes us to reject H_0?

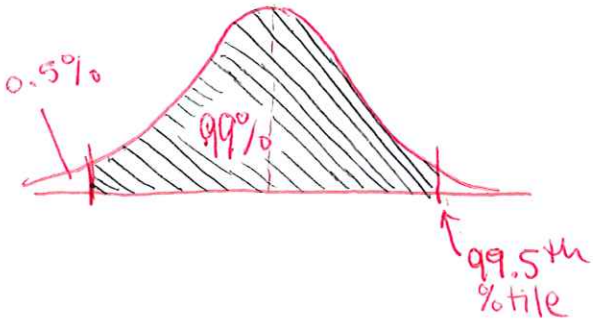


We know we'll get $\hat{\beta}_1$ values within 2.05 sd of the true β_1 value 95% of the time (for a t dist w/ df = 28), so anything further away must have probability less than 5%, ie. a p value less than 5%, which would result in us rejecting H_0 at a level of $\alpha = 0.05$.

reject H_0 if $\dfrac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{\hat{\sigma}^2}{(n-1)S_x^2}}} > 2.05$ or < -2.05

What if the rejection threshold for our test were $\alpha = 0.01$?

if $\alpha = 0.01$



$qt(0.995, df = 28) \approx 2.76$

Looking up the 99.5th percentile of a t dist w/ df = 28, we find that there is 99% prob. of getting an estimate $\hat{\beta}_1$ within $\pm$ 2.76 (estimated) standard

deviations of the true β_1 value. The chance of getting an estimated value at least 2.76 sd away is 1%. So if we got an estimated $\hat{\beta}_1$ more than 2.76 sd away from 0, we would reject the null hypothesis that $\beta_1 = 0$ at the $\alpha = 0.01$ level.

@ $\alpha = 0.01$: reject H_0 if $\dfrac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)} > 2.76$ or -2.76

Depending on our research question, it might make sense to use a one-sided alternative hypothesis in our test about the true slope.

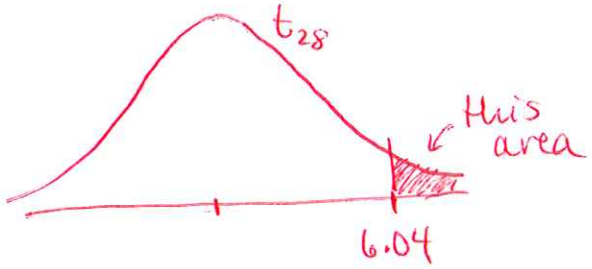
Can also do 1-sided test

Maybe we're specifically interested in the alternative that X and Y have a positive association.

eg $H_0: \beta_1 = 0$ vs. $H_a: \beta_1 > 0$

In that case, since our observed data produced an estimated slope of 16.9, we want the probability of getting a value at least that large. In other words, the prob. of a result at least 6.04 standard deviations above the hypothesized value.

$$P\left(\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{\hat{\sigma}^2}{(n-1)S_x^2}}} \geq 6.05 \right)$$



1 - pt(6.04, df = 28)

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hypothesis tests in R: examples in notebook